SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEM

WINTER SEMESTER 2025 – 2026

PROJECT TITLE:

PAYMENT SYSTEM

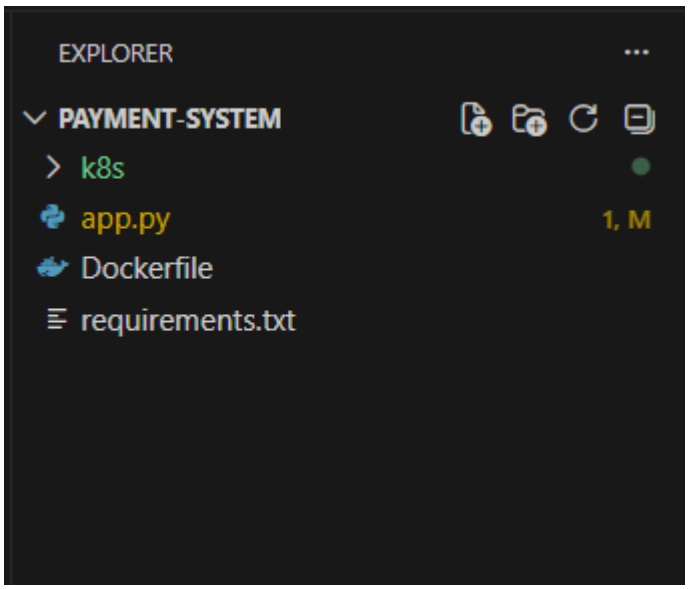
NAME: SAARUKESH PB

REG NO: 23MIS0341

FALCULTY NAME: Proff. RAMALINGAM M

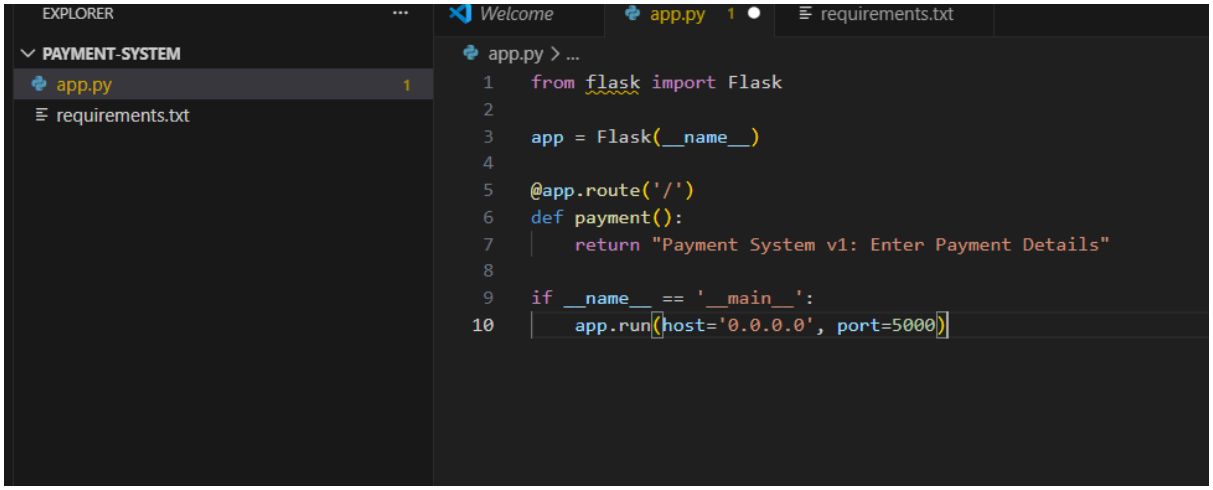
SLOT: C2 + TC2

STEP 1: Create the Folder Named **“PAYMENT-SYSTEM”**



VERSION 1:

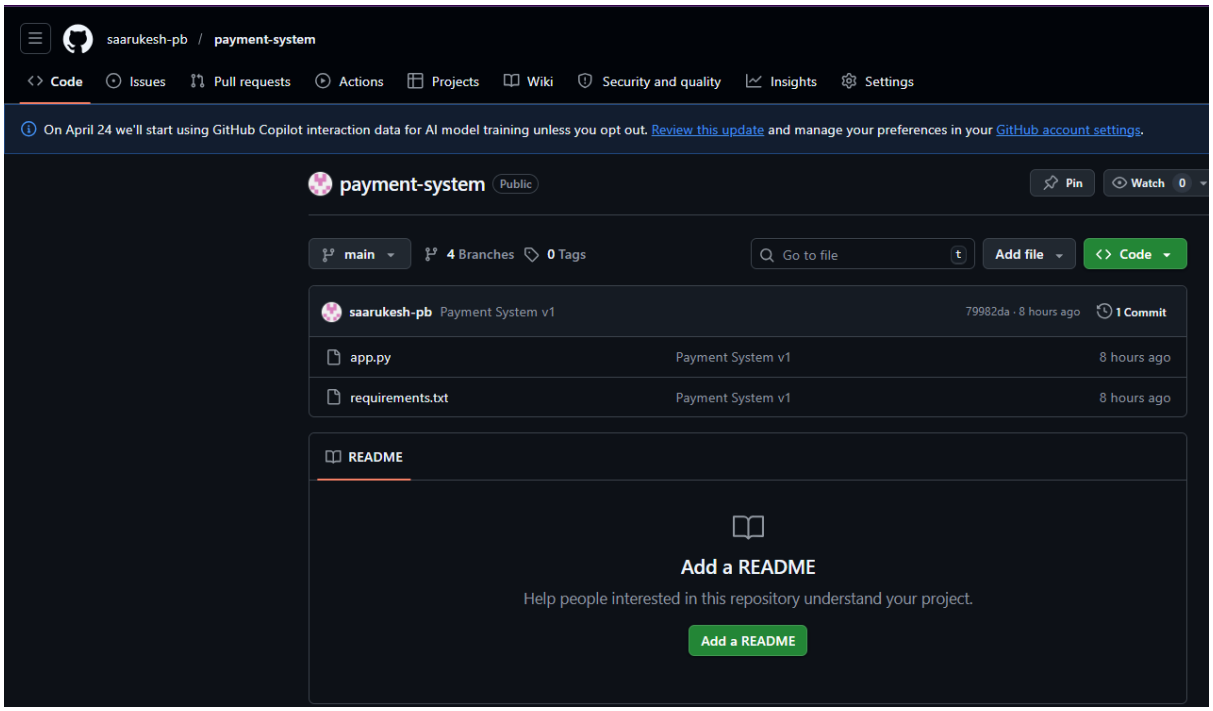
App.py



Create GitHub Repository And Push It:

```
git init
git add .
git commit -m "Payment System v1"
```

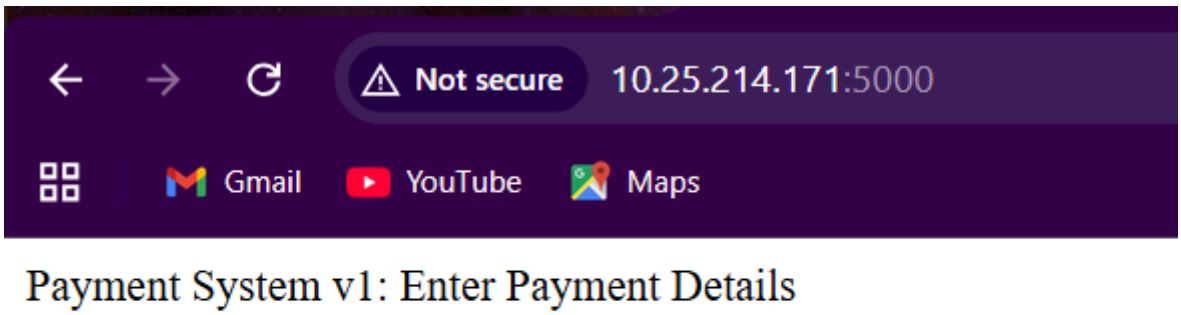
git remote add origin https://github.com/saarukesh-pb/payment-system.git
git push -u origin main



Create Branch for v1

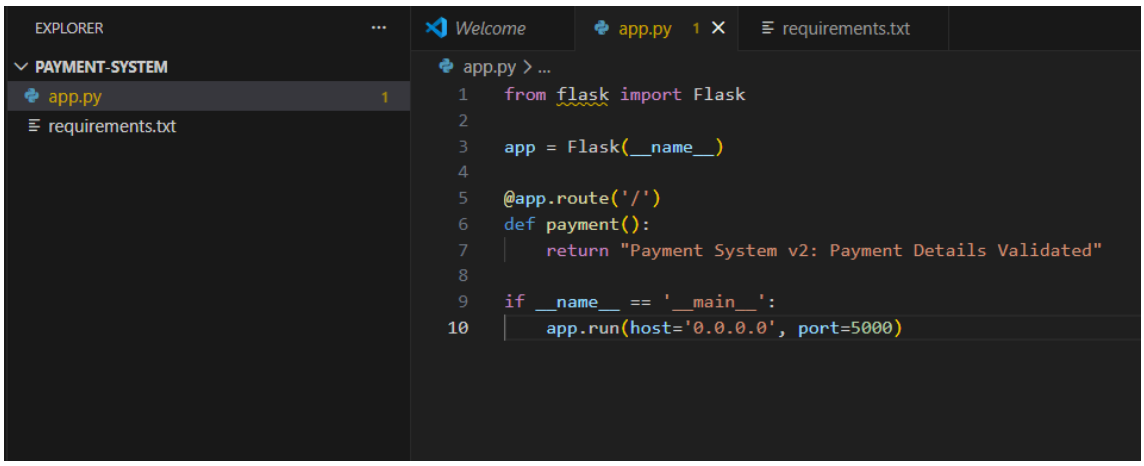
git checkout -b v1
git push origin v1

OUTPUT:



VERSION 2:

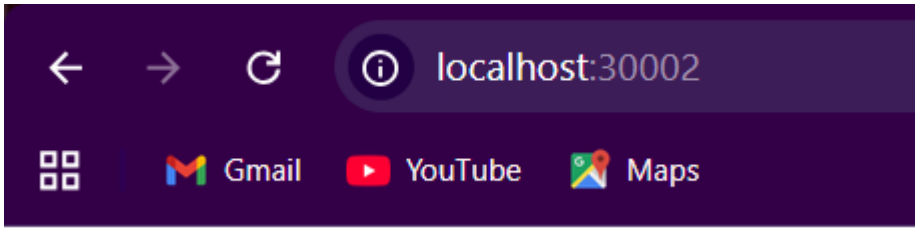
Update the App.py



Commit:

```
git add .
git commit -m "Payment System v2"
git push origin v2
```

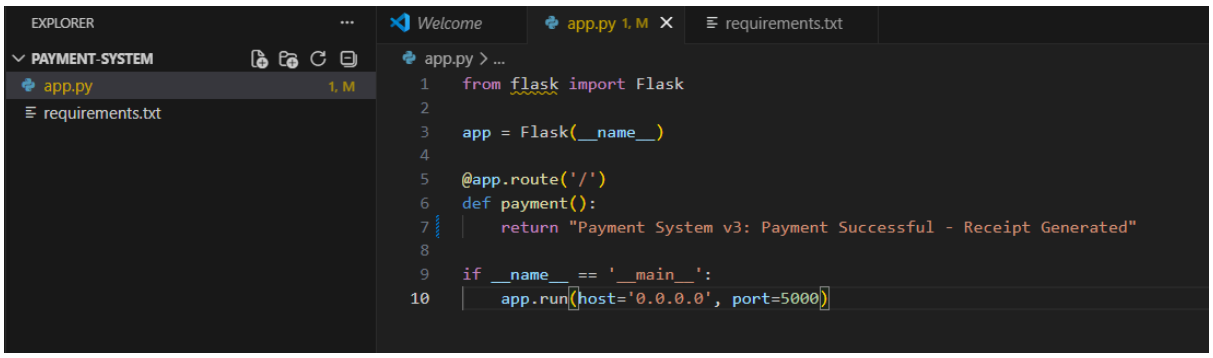
OUTPUT:



Payment System v2: Payment Details Validated

VERSION 3:

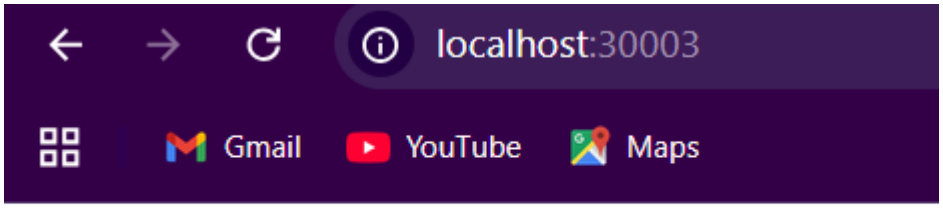
Update the app.py



Commit:

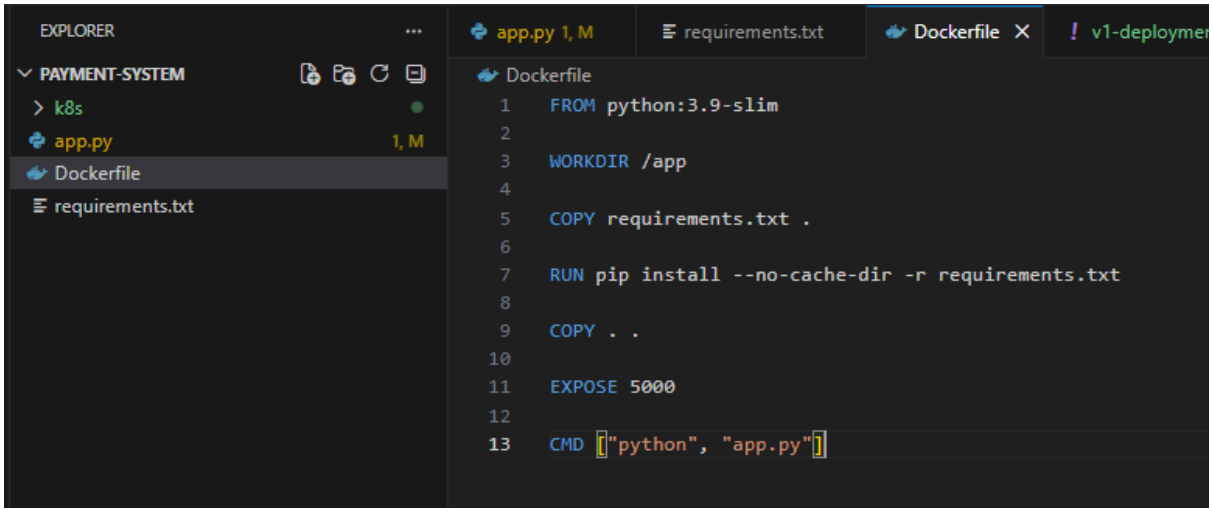
```
git add .
git commit -m "Payment System v3"
git push origin v3
```

OUTPUT:



Payment System v3: Payment Successful - Receipt Generated

Create Dockerfile



Build Docker Images:

For v1:

docker build -t payment-system:v1 .

```
PS C:\payment-system> docker build -t payment-system:v1 .
[+] Building 6.4s (11/11) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 211B
 => [internal] load metadata for docker.io/library/python:3.9-slim
 => [auth] library/python:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
 => [internal] load build context
 => => transferring context: 8.01kB
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY requirements.txt .
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:7b9af9b3c25b8457df8cfbc7c7ddf928958f71264592ddafce275e6927be220f
 => => exporting config sha256:1e2bb2409dd92e1db6f7290a41c8d72b8e6c84e8bb8d15d2ee2453676409d092
 => => exporting attestation manifest sha256:6e350473a8427debb085d7680da00efec9564c36cdeac249c071809f65de82bc
 => => exporting manifest list sha256:ed22b26134471dcec22fbf32a44f161e09c1c0829ef57c1f025e37eb80056396
 => => naming to docker.io/library/payment-system:v1
 => => unpacking to docker.io/library/payment-system:v1
```

For v2:

docker build -t payment-system:v2 .

```
PS C:\payment-system> docker build -t payment-system:v2 .
[+] Building 3.7s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 211B
 => [internal] load metadata for docker.io/library/python:3.9-slim
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
 => [internal] load build context
 => => transferring context: 4.32kB
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY requirements.txt .
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => CACHED [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:7b9af9b3c25b8457df8cfbc7c7ddf928958f71264592ddafce275e6927be220f
 => => exporting config sha256:1e2bb2409dd92e1db6f7290a41c8d72b8e6c84e8bb8d15d2ee2453676409d092
 => => exporting attestation manifest sha256:91b07957c6e32f1a43c3422c3620c58a30a7edad350dea3043c02ea253666f38
 => => exporting manifest list sha256:c09e2989c546713855d5435a8acda33ef27e400aeb87eea613bc459bf21e18b8
 => => naming to docker.io/library/payment-system:v2
 => => unpacking to docker.io/library/payment-system:v2
PS C:\payment-system>
```

For v3:

docker build -t payment-system:v3 .

```
PS C:\payment-system> docker build -t payment-system:v3 .
[+] Building 2.0s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 211B
 => [internal] load metadata for docker.io/library/python:3.9-slim
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
 => [internal] load build context
 => => transferring context: 4.32kB
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY requirements.txt .
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => CACHED [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:7b9af9b3c25b8457df8cfbc7c7ddf928958f71264592ddafce275e6927be220f
 => => exporting config sha256:1e2bb2409dd92e1db6f7290a41c8d72b8e6c84e8bb8d15d2ee2453676409d092
 => => exporting attestation manifest sha256:080878d001eacc0c629f40b47f6100391d375ca9478aaab586a043f2a041995f
 => => exporting manifest list sha256:0c5bf9d86f2ce4a3537138008dbd3645e14c755b2adcf2d2de55e0225b897aab
 => => naming to docker.io/library/payment-system:v3
 => => unpacking to docker.io/library/payment-system:v3
PS C:\payment-system>
```

Push to DockerHub

Tag images:

docker tag payment-system:v1 username/payment-system:v1
docker tag payment-system:v2 username/payment-system:v2
docker tag payment-system:v3 username/payment-system:v3

Setup Jenkins Pipeline:

Create **New item** Named "**payment-system-pipeline"**

Select **Pipeline** and click OK

Pipeline Script:

```
pipeline {
   agent any

   stages {

      stage('Clone Repository') {
         steps {
            git branch: 'v1', url: 'https://github.com/saarukesh-pb/payment-
system.git'
         }
      }

      stage('Build Docker Image') {
         steps {
            bat 'docker build -t saarukesh/payment-system:v1 .'
         }
```

```
        }

        stage('Push Docker Image') {
            steps {
                bat 'docker push saarukesh/payment-system:v1'
            }
        }

        stage('Run Container') {
            steps {
                bat '''
                docker rm -f payment-v1 || exit 0
                docker run -d --name payment-v1 -p 30001:5000 saarukesh/payment-
system:v1
                '''
            }
        }

    }
}
```

Click save and Build

Jenkins / payment-system-pipeline / #3

```
8736299eaeac: Layer already exists
b3ec39b36ae8: Layer already exists
2c74ce057189: Waiting
ab0e23e26166: Waiting
2c74ce057189: Waiting
ab0e23e26166: Waiting
2c74ce057189: Waiting
2c74ce057189: Pushed
ab0e23e26166: Pushed
v1: digest: sha256:1437185a1c85b1d53415de7c57276c8ef298014b17d5ebf7e4b28513236e6c8b size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Run Container)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\payment-system-pipeline>docker rm -f payment-v1   || exit 0
payment-v1

C:\ProgramData\Jenkins\.jenkins\workspace\payment-system-pipeline>docker run -d --name payment-v1 -p 30001:5000 saarukesh/payment-system:v1
33acd2c855a7556b9375dd49a7e230439d656446bdf0944571e154b7d22310b1
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins / payment-system-pipeline

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Favorite
- Open Blue Ocean
- Stages
- Rename
- Pipeline Syntax

Builds

Filter /

Today

- #3 1:00 PM
- #2 12:48 PM
- #1 12:38 PM

payment-system-pipeline

Stage View

	Clone Repository	Build Docker Image	Push Docker Image	Run Container
Average stage times: (full run time: ~17s)	3s	15s	2min 8s	1s
#3 Apr 12 13:00 No Changes	1s	4s	7s	2s
#2 Apr 12 12:48 1 commit	3s	37s	6min 18s	1s failed
#1 Apr 12 12:38 No Changes	5s	3s failed	169ms failed	84ms failed

Permalinks

- Last build (#3), 1 min 23 sec ago
- Last stable build (#3), 1 min 23 sec ago
- Last successful build (#3), 1 min 23 sec ago
- Last failed build (#2), 13 min ago
- Last unsuccessful build (#2), 13 min ago
- Last completed build (#3), 1 min 23 sec ago

Create Kubernetes Deployment:

VERSION 1

EXPLORER

PAYMENT-SYSTEM

- k8s
  - v1-deployment.yaml U
  - v1-service.yaml U
  - v2-deployment.yaml U
  - v2-service.yaml U
  - v3-deployment.yaml U
  - v3-service.yaml U
- app.py 1, M
- Dockerfile
- requirements.txt

app.py 1, M | requirements.txt | Dockerfile | v1-deplo

k8s > v1-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: payment-v1
spec:
  replicas: 1
  selector:
    matchLabels:
      app: payment-v1
  template:
    metadata:
      labels:
        app: payment-v1
    spec:
      containers:
      - name: payment-v1
        image: saarukesh/payment-system:v1
        imagePullPolicy: IfNotPresent
        ports:
        - containerPort: 5000
```

EXPLORER

PAYMENT-SYSTEM

- k8s
  - v1-deployment.yaml U
  - v1-service.yaml U
  - v2-deployment.yaml U
  - v2-service.yaml U
  - v3-deployment.yaml U
  - v3-service.yaml U
- app.py 1, M
- Dockerfile
- requirements.txt

app.py 1, M | requirements.txt | Doc

k8s > v1-service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: payment-v1-service
spec:
  type: NodePort
  selector:
    app: payment-v1
  ports:
    - port: 5000
      targetPort: 5000
      nodePort: 30001
```

VERSION 2

EXPLORER

PAYMENT-SYSTEM

- k8s
  - v1-deployment.yaml U
  - v1-service.yaml U
  - v2-deployment.yaml U
  - v2-service.yaml U
  - v3-deployment.yaml U
  - v3-service.yaml U
- app.py 1, M
- Dockerfile
- requirements.txt

app.py 1, M | requirements.txt | Dockerfile | v1-deployment.yam

k8s > v2-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: payment-v2
spec:
  replicas: 1
  selector:
    matchLabels:
      app: payment-v2
  template:
    metadata:
      labels:
        app: payment-v2
    spec:
      containers:
      - name: payment-v2
        image: saarukesh/payment-system:v2
        ports:
        - containerPort: 5000
```

EXPLORER

PAYMENT-SYSTEM

- k8s
  - v1-deployment.yaml U
  - v1-service.yaml U
  - v2-deployment.yaml U
  - v2-service.yaml U
  - v3-deployment.yaml U
  - v3-service.yaml U
- app.py 1, M
- Dockerfile
- requirements.txt

app.py 1, M | requirements.txt | Docke

k8s > v2-service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: payment-v2-service
spec:
  type: NodePort
  selector:
    app: payment-v2
  ports:
    - port: 5000
      targetPort: 5000
      nodePort: 30002
```

VERSION 3

EXPLORER

PAYMENT-SYSTEM

- k8s
  - v1-deployment.yaml U
  - v1-service.yaml U
  - v2-deployment.yaml U
  - v2-service.yaml U
  - v3-deployment.yaml U
  - v3-service.yaml U
- app.py 1, M
- Dockerfile
- requirements.txt

app.py 1, M | requirements.txt | Dockerfile | v1-dep

k8s > v3-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: payment-v3
spec:
  replicas: 1
  selector:
    matchLabels:
      app: payment-v3
  template:
    metadata:
      labels:
        app: payment-v3
    spec:
      containers:
      - name: payment-v3
        image: saarukesh/payment-system:v3
        ports:
        - containerPort: 5000
```

EXPLORER

PAYMENT-SYSTEM

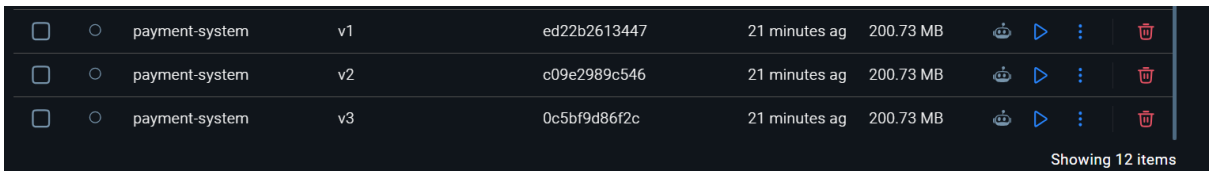
- k8s
  - v1-deployment.yaml U
  - v1-service.yaml U
  - v2-deployment.yaml U
  - v2-service.yaml U
  - v3-deployment.yaml U
  - v3-service.yaml U
- app.py 1, M
- Dockerfile
- requirements.txt

irements.txt | Dockerfile | v1-deployme

k8s > v3-service.yaml

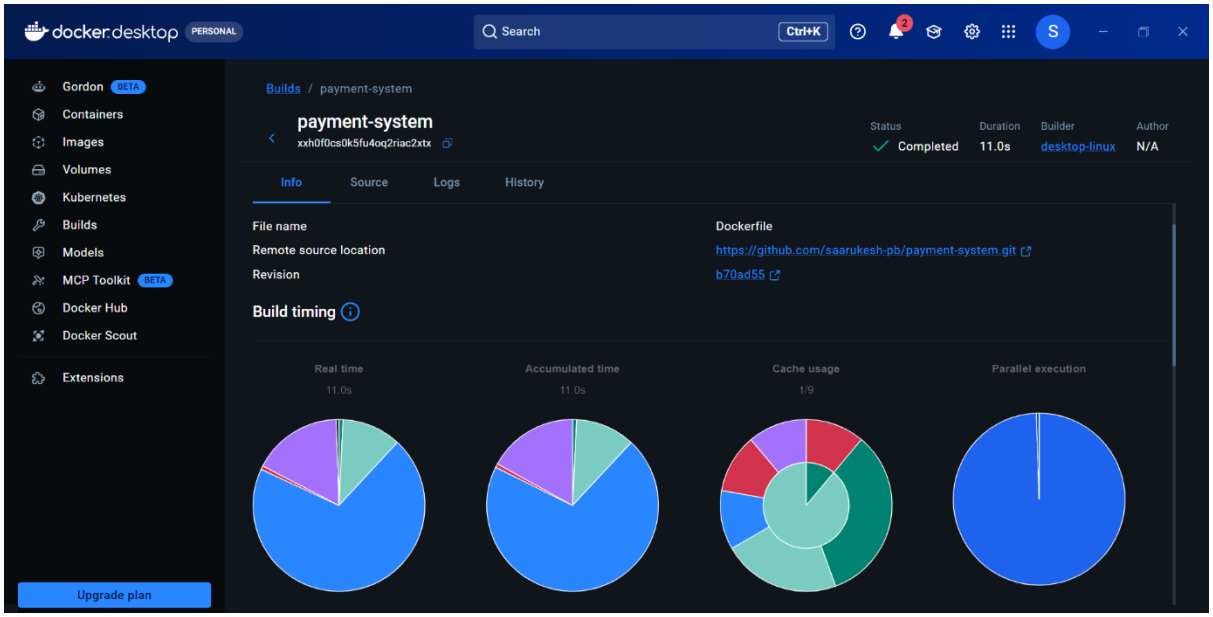
```yaml
apiVersion: v1
kind: Service
metadata:
  name: payment-v3-service
spec:
  type: NodePort
  selector:
    app: payment-v3
  ports:
    - port: 5000
      targetPort: 5000
      nodePort: 30003
```

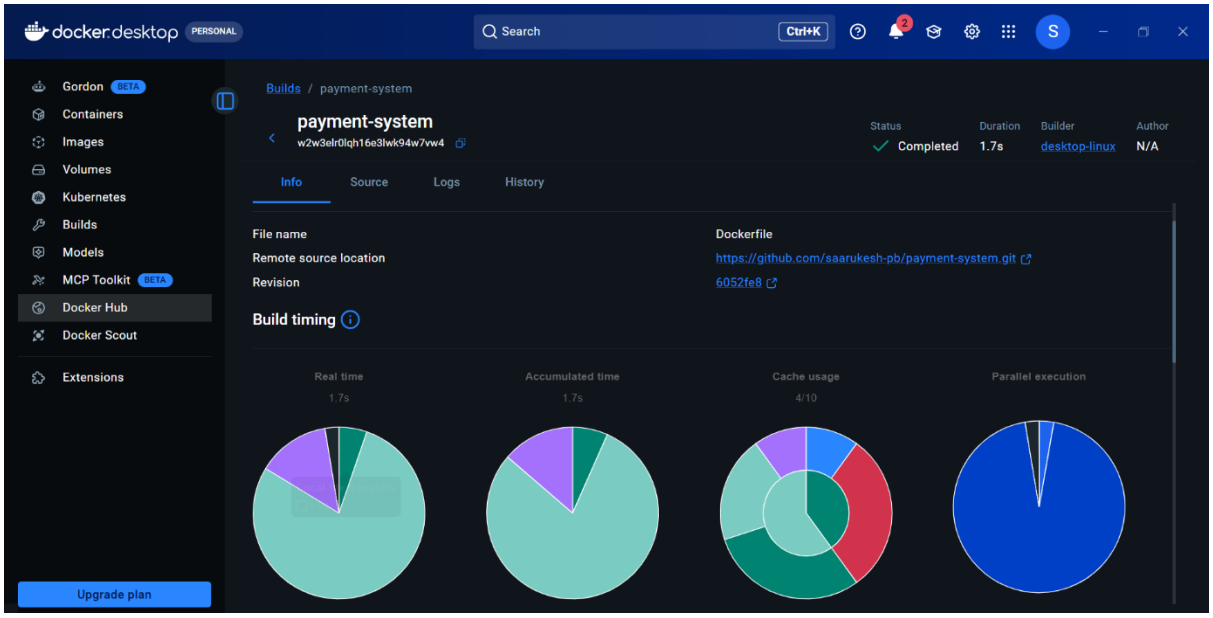
In the Docker Desktop:



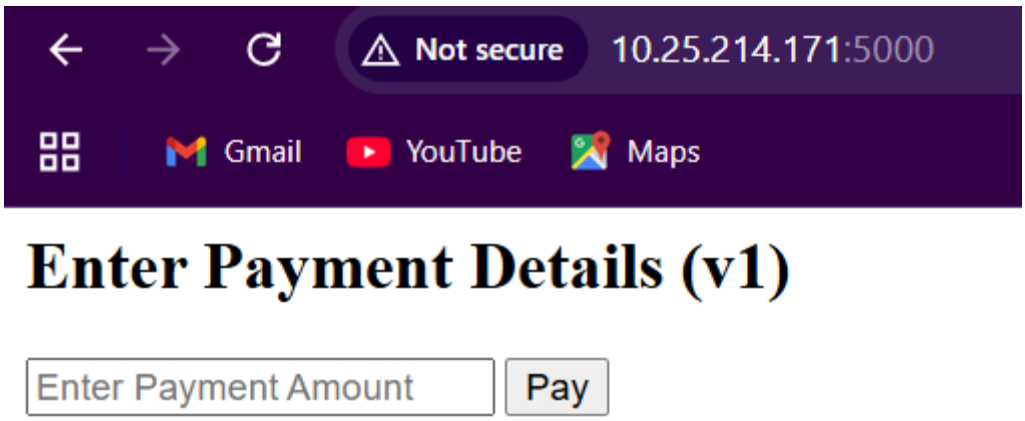
payment-system	v1	ed22b2613447	21 minutes ag	200.73 MB
payment-system	v2	c09e2989c546	21 minutes ag	200.73 MB
payment-system	v3	0c5bf9d86f2c	21 minutes ag	200.73 MB

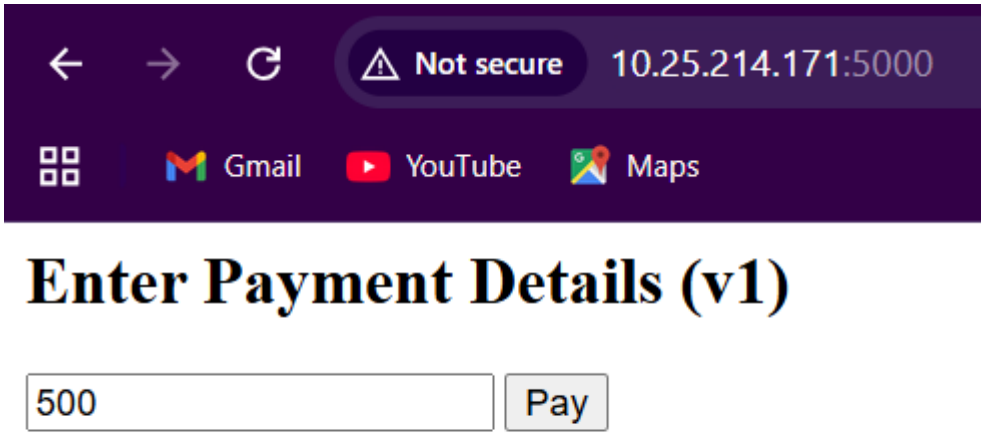
Showing 12 items





RESULT:


Not secure
10.25.214.171:5000
Gmail
YouTube
Maps
Enter Payment Details (v1)
Enter Payment Amount
Pay


Not secure
10.25.214.171:5000
Gmail
YouTube
Maps
Enter Payment Details (v1)
500
Pay

Not secure 10.25.214.171:5000/pay

Gmail YouTube Maps

Payment Validated (v2)

Amount: 500

Proceed to Receipt

Not secure 10.25.214.171:5000/receipt

Gmail YouTube Maps

Payment Successful (V3)

Transaction ID: 12345

Thank You for Your Payment!

Make Another Payment